Tiebreaker: The App That Ends the 'Where to Eat?' Debate

Fadi Mardelli
KU Leuven – Group T
Electronics and ICT Engineering

Nazar Kandiel
KU Leuven – Group T
Electronics and ICT Engineering

George-Adelin Aniței
KU Leuven – Group T
Electronics and ICT Engineering

Ezzdinne Belkaid
KU Leuven – Group T
Electronics and ICT Engineering

Jaskaran Singh Sidhu
KU Leuven – Group T
Electronics and ICT Engineering

Dele Ayeni
KU Leuven – Group T
Electronics and ICT Engineering

Abstract

Group decision-making for activities like dining out often encounters challenges such as uneven participation and indecisiveness, detracting from the collaborative experience. This study introduces Tiebreaker, a mobile application designed to enhance group dining decisions through an intuitive interface that integrates multiple motivational design principles, including the Fogg Behavior Model, Gamification, and Social Persuasion, amongst others.

Formative user testing informed iterative design improvements, while summative testing with five participants evaluated the app's usability and engagement. Key metrics including the System Usability Scale (SUS) and User Experience Scale (UES), demonstrated high user satisfaction and ease of use. Features such as progress indicators and social nudges were particularly effective in fostering engagement.

These findings highlight Tiebreaker's potential to streamline group decision-making while promoting inclusivity and enjoyment, with implications for future enhancements and broader applications.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**.

Keywords

Group decision-making, Motivational design, Fogg Behavior Model, Gamification, System Usability Scale (SUS), User Experience Scale (UES)

ACM Reference Format:
Fadi Mardelli, Nazar Kandiel, George-Adelin Aniței, Ezzdinne Belkaid, Jaskaran Singh Sidhu, and Dele Ayeni. 2025. Tiebreaker: The App That Ends the 'Where to Eat?' Debate. In *Proceedings of ACM Special Interest Group Conference (ACM SIGCONF).* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/1234567.1234568

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ACM SIGCONF, Location
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 123-4567-24-567/08/06
https://doi.org/10.1145/1234567.1234568

1 Introduction

1.1 Warrant for the Research

Group decision-making is an essential component of social interaction, necessary in numerous aspects of life, such as teamwork in workplaces, spending time with family, and engaging with personal friend groups or broader social connections. Dining out, in particular, is one of the more popular team-building activities in professional settings, enhancing collaboration and camaraderie among colleagues. Similarly, it is a common way to celebrate important milestones with family or friends, making it a crucial element of shared social experiences.

Given its significance, it is vital that group dining experiences are smooth and enjoyable for all participants. While several apps exist to recommend or filter restaurants based on individual preferences, most fail to address the complexities of decision-making in larger groups. The apps that do attempt to account for group preferences often fail to streamline the decision-making process due to their complexity, lack of intuitive design, and features that inadvertently add more effort rather than simplifying the experience. This gap highlights the need for tools designed specifically to support collaborative and inclusive decision-making.

To address this, *Tiebreaker* was developed as a mobile application focused on simplifying group dining decisions. By aggregating user preferences and incorporating motivational design elements, Tiebreaker ensures that decision-making is both efficient and enjoyable. This research explores the app's ability to enhance group experiences through a user-friendly and engaging platform.

1.2 Problem Statement

Within a group, individuals may hesitate to express their opinions, fearing they might slow down the decision-making process or disrupt group harmony. Others may dominate the discussion due to their naturally vocal or opinionated nature, while some, despite having clear preferences, refrain from asserting their choices to avoid imposing their will on others.

In the context of group dining, these challenges are further amplified, as the goal is to foster enjoyment, connection, and engagement. However, the decision-making process often becomes cumbersome, creating unnecessary tension or even negative, argumentative energy. This undermines the purpose of dining together and can leave participants feeling dissatisfied.

To address these issues, *Tiebreaker* must not only streamline the process and ensure fair consideration of all preferences but also motivate users to rely on it consistently. The app would provide a more efficient, engaging, and enjoyable experience than relying

on traditional group discussions to make decisions. By promoting inclusivity and reducing friction, this approach preserves the collaborative and enjoyable spirit of group dining while encouraging continued use of the application due to its effectiveness in picking the restaurant that matches the group members' preferences.

1.3 Research Questions

(1) How intuitive and user-friendly is Tiebreaker for diverse groups during the group dining decision-making process?
(2) How effective are Tiebreaker's design elements (e.g., quizzes, progress indicators, and social nudges) in motivating user engagement and improving the efficiency and satisfaction of group dining decisions compared to traditional discussions?
(3) What features of Tiebreaker encourage users to consistently rely on the app for group dining decisions instead of returning to traditional methods?

2 Background and Related Work

2.1 Information on the Application Domain

Group dining decisions are a common social experience, whether among friends, coworkers, or families. Despite their apparent simplicity, these decisions can become challenging as individual preferences, such as dietary restrictions, budget constraints, and taste profiles collide. When multiple people collaborate on a choice, traditional discussion methods often involve lengthy back-and-forth exchanges, uncertainty about who should make the final call, and concerns that certain voices may dominate while others remain unheard. This dynamic underscores the importance of tools that reduce friction and promote inclusivity.

Recent work has highlighted the complexities of group decision-making processes, demonstrating how easily negotiations can break down or become dominated by particular individuals [2]. Some participants may be reluctant to express strong preferences, while others push aggressively for their favored options, leading to rushed or suboptimal outcomes. Within the dining domain, this problem is further increased by the wide variety of cuisines, dietary requirements, and cost considerations. Consequently, there remains a clear opportunity for a specialized application that both balances efficiency and fairness, and offers a seamless, enjoyable experience to foster greater participation and consensus-building among group members.

2.2 Theories on Motivation

Designing an engaging and persuasive system often involves applying a well-established framework, such as the Fogg Behavior Model (FBM), which posits that behavior occurs when three core elements, Motivation, Ability, and Trigger converge at sufficient levels [3]. In the context of a group dining application, motivation can stem from a variety of factors: the desire to avoid conflict, the aspiration to share enjoyable meals, or simply the need to streamline an otherwise time-consuming process. Ability, on the other hand, emphasizes ease of use. If the interface is cumbersome or demands too many steps to input preferences, even highly motivated users may abandon the system. Reducing complexities, such as minimizing form fields, simplifying group setup, or automating routine tasks helps preserve user engagement.

Triggers are the cues that remind or prompt users to take action at the appropriate moment. Within a group dining app, these might take the form of notifications prompting members to finalize their choices, progress bars showing how close the group is to reaching consensus, or mild social nudges indicating who is "ready" or "not ready." Such triggers must strike a delicate balance: they should be timely enough to prevent stalling, yet they must respect users' autonomy and sense of comfort. Research on technology-mediated nudging underscores the diverse ways designers can leverage triggers to guide user behavior subtly [1]. By carefully orchestrating motivation, ability, and triggers, a group dining solution can maintain consistent involvement, reduce stress, and ensure that each group decision is both efficient and harmonious.

2.3 State of the Art (SOTA)

Existing solutions for collective dining decisions generally tackle either preference aggregation or convenience but rarely address the motivational underpinnings of the decision-making process. On one end, some apps manage to compile basic user preferences for cuisines and dietary requirements but offer minimal support for negotiation and consensus-building. At the same time, research on group behavior has shown that conflicts can arise not only from conflicting tastes but also from deeper motivational factors [4]. When certain members are more assertive or when others withdraw from active participation, the group's dynamics can swiftly deteriorate, leading to poorer outcomes or prolonged stalemates.

Moreover, while a few group recommender prototypes have integrated functions like voting or rating, these tend to work best with homogeneous groups and can struggle with heterogeneity. Gamification elements, such as earning badges or tracking progress, have been proposed to keep people engaged, yet these can also produce unintended consequences e.g., intensifying competitive behaviors or demotivating those who do not see value in rewards [4]. Additionally, insights from focus group studies reveal that users often expect more than mere preference pooling; they seek guidance in coordinating viewpoints, resolving potential conflicts, and ensuring that everyone's voice is heard [2].

Against this backdrop, Tiebreaker aims to fill the gap by merging preference aggregation with a motivational design that aligns with Fogg's model. Unlike apps that only highlight average preferences or generate simplistic aggregated scores, Tiebreaker emphasizes real-time persuasion elements, like subtle reminders, progress indicators, and private or public nudges, to keep all participants involved. In doing so, it addresses the potential pitfalls of one-size-fits-all recommendation mechanisms, offering a more contextual, user-centric, and socially aware system. Ultimately, few existing platforms seamlessly integrate collaboration features, nudging strategies, and a robust motivational backbone, which positions Tiebreaker as a promising new approach in the state of the art for group dining applications.

3 Design and Development

The design and development of *Tiebreaker* followed a user-centered approach, leveraging iterative feedback from both low-fidelity (Lo-Fi) and high-fidelity (Hi-Fi) prototyping phases. The goal was to create an intuitive, engaging application that streamlines group

decision-making by combining user preferences. This section outlines the co-design process, user requirements, prototype development, and the motivational elements embedded within the final product.

3.1 Co-design Process

The co-design process for *Tiebreaker* combined internal brainstorming and external user feedback to address challenges in group dining decisions. Early brainstorming sessions identified common pain points, particularly the difficulty of reaching a consensus when selecting restaurants, leading to the conceptualization of an app designed to streamline this process.

To ensure the design aligned with user needs, the team developed personas representing different dining behaviors and group dynamics. These personas informed the app's flow, aligning features with user expectations to create an intuitive and inclusive experience.

The design phase involved collaborative wireframing in Figma, with team members iterating on the app's structure and functionality in real-time. Regular reviews allowed for adjustments that resolved friction points, resulting in the final low-fidelity (Lo-Fi) prototype used for user testing.

A similar process guided the high-fidelity (Hi-Fi) prototype, where one team member developed a cohesive design system while others provided feedback to refine the visual and interactive elements. This iterative approach ensured the final prototype balanced functionality with an engaging, polished user experience.

3.2 User Requirements

Tiebreaker's user requirements were shaped by two key personas: the group leader and the group member. These roles captured the dynamics of group dining, ensuring the app addressed real user needs.

Group leaders initiated group creation, invited participants, and finalized restaurant selection, while group members joined existing groups and contributed preferences. The app's flow adjusted slightly to reflect these roles, with all users landing on the "Members" page post-formation. The primary difference was the group leader's access to the "Generate" button, used to finalize decisions, while the rest of the experience remained uniform for all users.

Tiebreaker was designed to tackle frustrations faced by both personas. Group leaders often managed the burden of decision-making, while members risked feeling excluded from the process. By automating restaurant selection through collective preferences, the app reduced delays, minimized conflict, and created a smoother, more inclusive experience for all participants.

3.3 Lo-Fi Prototype

The development of the Lo-Fi prototype was a critical step in bringing the app's conceptual framework to life. These prototypes mapped out the essential screens and interactions, outlining the process from group creation to restaurant selection. The flow of the app centered on minimizing friction points, ensuring that users could intuitively move through stages without unnecessary complexity.

As envisioned, the app's journey began with a group leader creating a session and sharing a unique code with participants. Group

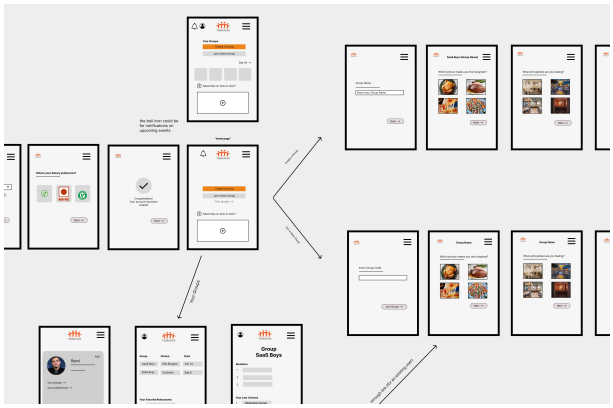


Figure 1: Low Fidelity Prototype

members entered the code to join the session, after which both leaders and participants completed preference quizzes that informed the app's restaurant matching algorithm. Upon the completion of all quizzes, the group leader triggered the generation of a restaurant recommendation, with the result displaying comprehensive details such as the restaurant's contact information, directions, website, and images.

The prototype effectively captured the intended user journey, providing a tangible representation of the app's core functionality. However, the transition from conceptual design to practical implementation necessitated rigorous user testing to validate assumptions and refine areas of friction.

3.4 Formative User Test Learnings

User testing with university students, Tiebreaker's target demographic, was conducted to simulate real-world group dining scenarios. Sessions involved two participants interacting with the app together, reflecting the collaborative nature of the app. Each session lasted about fifteen minutes, during which users explored core features and discussed their decisions.

Participants generally found the app intuitive, with logical screen transitions and actions. However, hesitation was observed after group creation or joining, as users were initially prompted to set dining preferences immediately. This step felt intrusive, with users preferring to set preferences later.

In response, the app's flow was adjusted to allow users to complete preferences at their convenience, creating a smoother experience that better aligned with user behavior. This non-linear approach improved flexibility and usability, ensuring the app adapted to different group dynamics. The iterative refinements reflect Tiebreaker's focus on user-centered design, balancing structured guidance with the freedom for users to interact at their own pace.

3.4.1 Hi-Fi Prototype. The high-fidelity (Hi-Fi) prototype built upon the foundation established during the low-fidelity (Lo-Fi) phase by incorporating polished visual elements, dynamic interactions, and enhanced user engagement features. This stage of the design process focused on refining the app's aesthetic appeal while ensuring that the overall experience remained intuitive and aligned

with user needs. The visual language of the Hi-Fi prototype emphasized warmth, invitingness, and simplicity, achieved through the use of vibrant and inviting color palettes, smooth transitions, and engaging animations that guided users through each stage of the app.

One of the primary objectives of the Hi-Fi phase was to introduce motivational and interactive features designed to sustain user interest and enhance overall satisfaction. Badges were implemented to foster a sense of achievement and reward users for actions such as creating groups, participating in events, and maintaining consistent engagement with the app. This system of Gamification and social validation not only reinforced positive behavior but also contributed to the app's sense of community and collaboration.

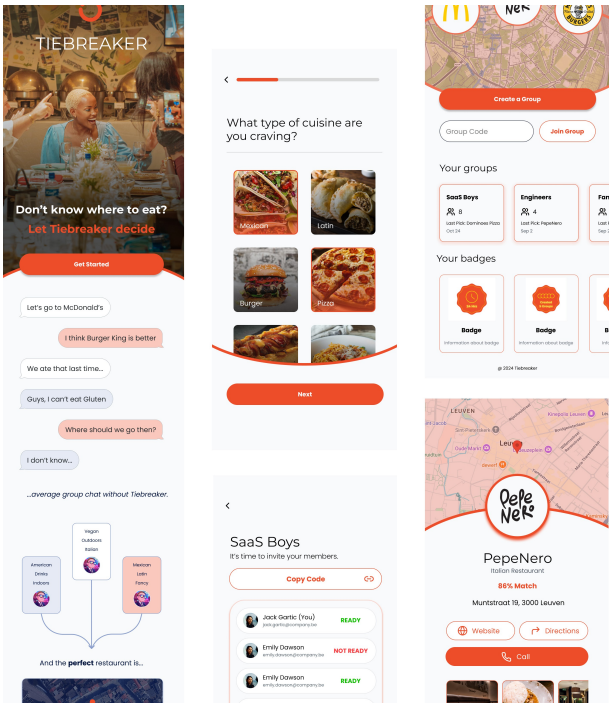


Figure 2: High Fidelity Prototype

A standout feature of the Hi-Fi design was the integration of a roulette animation to present the final restaurant selection. This approach introduced an element of excitement and unpredictability, tapping into the Octalysis Framework's core drive of unpredictability and curiosity. The roulette mechanic transformed a routine decision into a playful, memorable moment, reinforcing the app's goal of making group dining decisions enjoyable and engaging.

To support user motivation through gamification and provide clarity throughout the decision-making process, the prototype incorporated progress indicators, particularly within the preference quiz component. Progress bars offered real-time feedback, creating a sense of incremental achievement and encouraging users to complete tasks without feeling overwhelmed. Additionally, a subtle social persuasion mechanism was introduced through "Ready/Not Ready" status tags displayed alongside participant names on the page where all the group members are displayed. This non-intrusive feature encouraged engagement by allowing users to track group progress without feeling pressured to act immediately, preserving individual autonomy within the collaborative experience.

By synthesizing user feedback and incorporating dynamic, visually appealing features, the Hi-Fi prototype represented a significant step toward a market-ready product. The iterative, user-centered design approach ensured that Tiebreaker not only met functional requirements but also provided an engaging, enjoyable experience for its users.

4 Method

This section describes the process of summative user testing of the application including the following aspects: Participants, measurement instruments, and procedure.

4.1 Participants

For the summative user testing, five university students were invited to try the app and perform a set of given tasks. The students were selected based on availability and willingness to participate. As the students dine out frequently, they make up a sample that is representative of the target audience of the application.

4.2 Measurement Instruments

The instruments used for the user testing included a phone and two computers. These tools were essential for accessing the app and performing the tasks under controlled conditions.

4.3 Procedure

Human-computer interaction principles were used as a reference for planning the user testing. The process is outlined in three key steps:

4.3.1 Task Preparation. Five tasks were prepared for the users to assess different core functionalities of the application:

(1) Sign in to the app with the given credentials.
(2) Create a group with a specified name, time, and date.
(3) Invite a friend to the created group and generate results.
(4) Join a group using a given code.
(5) Change the profile avatar.

4.3.2 Task Hypothesis and Metrics. The hypothesis for each task was set with a specific time frame anticipated for completion:

- **Task 1:** 45 seconds
- **Task 2:** 30 seconds
- **Task 3:** 1 minute
- **Task 4:** 30 seconds
- **Task 5:** 30 seconds

The primary metric for evaluating user performance was the time taken to complete each task, focusing on efficiency given the app's mobile nature.

4.3.3 Feedback Collection. The System Usability Scale (SUS) and User Experience Scale (UES) were utilized to gather quantitative feedback, using Google Forms. These forms included 12 questions split into categories such as Engagement, Usability, Aesthetics, and Emotions. User responses were documented and will be analyzed in the results section.

The user testing sessions were conducted in a reserved room to minimize distractions. Each participant was given a brief presentation about the application and the tasks ahead, emphasizing that the testing focused on the app, not the user's abilities. During the sessions, users were encouraged to think aloud to provide real-time feedback and insights.

4.3.4 Post-Testing Questions and Feedback. After completing the tasks, users were asked specific questions about the motivational design elements of the app:

(1) What did you think about the spinning logos? Did they keep you interested or distract you?
(2) What did you think about the quiz using the progress bar and images instead of text?
(3) What did you think about the analysis loading page?

Feedback from these questions was noted, and participants were then asked to complete the SUS, UES, and Net Promoter Score (NPS) to quantitatively measure their overall experience and satisfaction.

4.4 Conclusion of Testing

Participants were thanked for their involvement, and all data collected were secured for further analysis, which is detailed in the results section of this paper.

5 Results

This section discusses the results of the user testing, including observations and feedback received. The performance of each task is detailed alongside the outcomes of the System Usability Scale (SUS), User Experience Scale (UES), and Net Promoter Score (NPS).

5.1 Task Performance

The average time taken to complete the tasks was as follows:

- **Task 1:** 66.86 seconds
- **Task 2:** 30.37 seconds
- **Task 3:** 101.34 seconds
- **Task 4:** 21.53 seconds
- **Task 5:** 42.37 seconds

These results are visualized in Figure 3, which provides a comparative analysis of actual performance against hypothesized values.

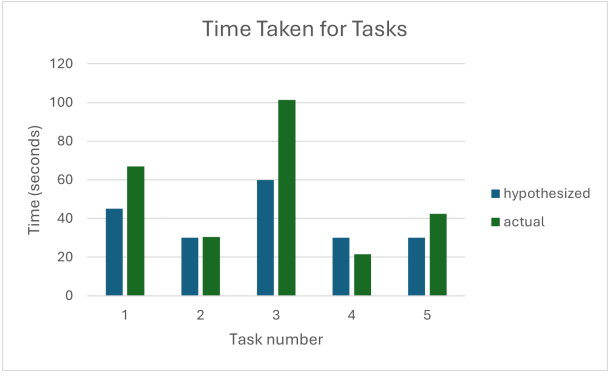


Figure 3: Summative User Testing Task Duration

The SUS followed a pattern of one positive and one negative question following each other, the results were separated into two segments, the positive segment and the negative segment.

For the positive segment, all questions were answered with agree or strongly agree. For the negative segment, all questions except one were answered disagree or strongly disagree, the exception answer being neutral by one response for the question: I often feel disconnected or uninterested while using the app. For UES, similar results as the SUS were obtained with two exceptions. The question about the overall layout and aesthetics received a unanimous response of strongly agree by all five participants. The question "could achieve my goals efficiently" received two strongly agree, one agree and two strongly disagree responses.

The NPS had an average score of 8.3. The answers to the motivational design question about the spinning logos were overall positive with some additional remarks such about the spinning logos as a nice feature but could feature local restaurants. The question regarding the progress bar received mixed responses, with two participants saying they didn't notice it and the rest saying it kept them engaged and motivated to finish the quiz.

The question about the analysis page got a positive response, the participants liked that they could see the progress of the calculation of the algorithm, and it made them excited to see the result. The customer support feature also received a positive response. Suggestions for improvements were, providing a link to share for the group instead of copying a code, mention that there is a limit of three cuisine choices in the quiz, and feedback when changing the avatar to avoid confusion.

6 Discussion

The results from summative user testing provided valuable insights into *Tiebreaker*'s usability, engagement, and motivational design, directly addressing the research questions posed at the start of development.

6.1 User-Friendliness and Intuitiveness

User testing consistently demonstrated that *Tiebreaker* is intuitive, easy to navigate, and quick to use, addressing the first research question regarding user-friendliness. Participants highlighted the app's simplicity and clear progression of tasks, with group creation completed in an average of 30.37 seconds. This efficient task completion reflected the app's minimal barriers to interaction and the logical flow from group creation to restaurant selection.

Formative user testing identified initial hesitation when users were prompted to set preferences immediately after joining a group. This feedback led to refinements in the Hi-Fi prototype, allowing users to complete preferences at their convenience. Summative testing confirmed the success of this adjustment, as users completed most tasks smoothly. However, Task 3—inviting friends and generating results—averaged 40 seconds longer than anticipated. Users expressed uncertainty about inviting participants, expecting in-app sharing options rather than copying codes manually. Despite this, the overall process was described as intuitive, with users emphasizing the app's practicality for real-world group dining scenarios.

A few minor usability issues surfaced during testing, including brief confusion between sign-in and sign-up screens and unclear

avatar customization processes, which led to longer-than-expected completion times for profile adjustments (42.37 seconds compared to the expected 30). These insights prompted further design refinements to improve feedback and clarity. Users expressed enthusiasm for the app's potential, with one participant stating, "I can see myself using this every week when going out with friends," reflecting a clear market need.

6.2 Effectiveness of Motivational Design Elements

Tiebreaker's design elements, such as quizzes, progress indicators, and social nudges, effectively promoted engagement and streamlined decision-making, addressing the second research question. Gamification features like badges, spinning logo animations, and progress bars enhanced user motivation by aligning with the Octalysis Framework and the Self-Determination Theory (SDT) principles of competence and autonomy.

Participants responded positively to the visual quiz format, with images providing a more engaging experience compared to traditional text-based options. Progress indicators encouraged task completion by offering users a clear sense of advancement, reinforcing motivation. While some participants reported not noticing the progress bar, those who did noted it contributed to their sense of accomplishment. One user remarked, *"Seeing how close I was to finishing the quiz kept me going."*

The roulette animation used for the final restaurant selection was well-received, introducing excitement and unpredictability, tapping into the Octalysis Framework's "unpredictability and curiosity" [REFERENCE HERE] drive. Users described the feature as fun and interactive, likening it to a wheel of fortune, which transformed a routine selection into an enjoyable experience.

6.3 Encouraging Long-Term Engagement

Addressing the third research question, *Tiebreaker*'s ability to automate restaurant selection through collective preferences emerged as a significant driver of long-term engagement. Users highlighted the app's efficiency in resolving common pain points associated with traditional dining decision-making. By combining preferences and generating unbiased results, the app minimized group conflict and streamlined the process, offering a structured alternative to manual coordination.

Gamified features such as badges and visual progress reinforced continued use, fostering a sense of achievement. Additionally, the spinning logo animations and interactive quizzes added entertainment value, distinguishing *Tiebreaker* from competitors. Users noted that *Tiebreaker*'s seamless blend of functionality and engagement made them more likely to return to the app for future group decisions.

While feedback was overwhelmingly positive, some users suggested expanding the restaurant database to include more local options and allowing for avatar uploads, reflecting areas for potential enhancement. Despite these suggestions, Tiebreaker's automated decision-making and motivational design positioned it as a compelling, user-centric alternative to traditional group dining coordination methods.

6.4 Ethical Considerations and Future Work

Throughout the development of *Tiebreaker*, the design team prioritized ethical engagement, deliberately avoiding dark patterns that could manipulate or frustrate users. The app's design allowed users full autonomy over their quiz participation, enabling them to skip or delay sections without penalty. This ensured that users never felt trapped in a continuous loop of engagement.

User privacy was another critical consideration. Dietary preferences and quiz responses were made visible only to individual users, safeguarding sensitive information from group members. This decision was based on ethical concerns about potential embarrassment or social pressure, which could arise if preferences were publicly displayed.

Despite the lack of overt dark patterns, the design team acknowledged the potential ethical implications of social nudges, such as the "Ready/Not Ready" indicator. While this feature promotes collaboration, the team plans to monitor user feedback to ensure it does not inadvertently lead to negative experiences.

6.5 Limitations and Future Directions

While *Tiebreaker* received positive feedback, certain constraints limit its initial release. A primary limitation is the geographic restriction of the app to Leuven, Belgium, resulting from technical dependencies on the Google Maps API. This prevented dynamic restaurant fetching outside the pre-set location. Expanding the reach of the app to additional cities and countries remains a priority to improve accessibility and scalability. In addition, technical challenges shaped aspects of the design, particularly the inability to implement dynamic map pins for selected restaurants. As a result, the app relied on static maps, reducing interactivity and visual feedback. Addressing this in future updates will create a more engaging and seamless user experience.

Future iterations of *Tiebreaker* will focus on expanding functionality and improving user engagement. A critical development goal is the integration of dynamic location input, allowing users to select dining locations rather than being restricted to a fixed city. This enhancement will significantly broaden the usability and reach of the app.

To simplify group formation, direct user invites will replace the reliance on manually entering group codes, allowing leaders to share links for faster and more efficient participation. In addition, expanded gamification elements, including new badges and interactive rewards, will build on the current framework to drive long-term engagement and user retention.

Accessibility remains a key focus for future development. Inclusion of multi-language support and features designed for visually impaired users will enhance inclusivity, ensuring *Tiebreaker* caters to a broader and more diverse demographic.

By addressing these limitations and continually refining the app based on user feedback, *Tiebreaker* aims to evolve into a comprehensive solution for group dining decisions, capable of serving users in different regions and improving the overall dining experience.

7 Conclusion

This study explored the vital question of how to improve group dining decisions through a dedicated mobile application, Tiebreaker.

Designed with meticulous attention to detail, Tiebreaker evolved through iterative testing phases, incorporating low- and high-fidelity prototypes to ensure a user-friendly and efficient interface. Our approach involved detailed summative user testing with university students, focusing on key usability metrics such as the System Usability Scale (SUS) and User Experience Scale (UES). The results were promising: Tiebreaker significantly streamlined the decision-making process for groups. Key features like progress indicators and social nudges didn't just simplify decisions—they made the experience more engaging and satisfying for users. Reflecting on these outcomes, it's clear that embedding motivational design principles in such tools can dramatically enhance user engagement and decision-making effectiveness. This study not only underscores the importance of user-centered design but also offers valuable insights that could be leveraged to enhance interactive tools for group activities across various domains.

8 Appendix

Link to Google forms: UES, SUS

Acknowledgments

We would like to express our sincere gratitude to Professor Vero Vanden Abeele, whose guidance and expertise were invaluable throughout the course of our Research and Development experience. Her insights and encouragement were instrumental in shaping both the theoretical and practical aspects of our SaaS application. We are also deeply thankful to all the technical staff and teaching assistants who provided us with support and assistance during the development process. Their technical expertise and timely advice helped us overcome many challenges we faced, and their contributions were critical to the success of our project. Lastly, we appreciate the collaborative efforts of all team members and peers who engaged in constructive discussions and shared their perspectives, further enriching our learning experience and project development.

References

[1] Ana Caraban, Evangelos Karapanos, Daniel Gonçalves, and Pedro Campos. 2019. 23 Ways to Nudge: A Review of Technology-Mediated Nudging in Human-Computer Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. doi:10.1145/3290605.3300733

[2] A. Delić, H. Emamgholizadeh, F. Ricci, and J. Masthoff. 2024. Supporting Group Decision-Making: Insights from a Focus Group Study. In *Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP '24)*. 301–306. doi:10.1145/3627043.3659538

[3] B. J. Fogg. 2009. A Behavior Model for Persuasive Design. Available at: https://www.demenzemedicinagenerale.net/images/mens-sana/Captology_Fogg_Behavior_Model.pdf.

[4] R. Van Roy and B. Zaman. 2019. Unravelling the Ambivalent Motivational Power of Gamification: A Basic Psychological Needs Perspective. *International Journal of Human-Computer Studies* 127 (Jul. 2019), 38–50. doi:10.1016/j.ijhcs.2018.04.009